Northern Counties Examination Board **Candidate copy**

General Certificate of Secondary Education

Computer Science – Paper 2: Programming and Algorithms

Thursday 4 June 2026 – Morning
Time allowed: 1 hour 30 minutes

Candidate name	Centre number	Candidate number

Instructions

- Use black ink or a black ball-point pen.
- Answer all questions in the spaces provided.
- Do all rough work in this book and cross through any work you do not want to be marked.
- Calculators are not permitted in this paper.

Information

- The total mark for this paper is 40.
- The marks for each question are shown in brackets.
- Program code is written in Python 3.

Read each question carefully before you start to write your answer. You are reminded of the need for clear presentation in your answers.

For Examiner's use only: Section A ____ / 18 Section B ____ / 22 Total ____ / 40

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Section A

Question 1

A program stores the daily rainfall, in millimetres, for one week in a list.

(a) State the data type that should be used for each rainfall value. Give a reason for your answer. **[2]**

(b) Explain the difference between a list and a constant. **[2]**

Study the following program.

```
rain = [12, 0, 3, 0, 25, 7, 0]
dry = 0
for day in rain:
    if day == 0:
        dry = dry + 1
print("Dry days:", dry)
print("Total:", sum(rain))
```

(c) Write down exactly what the program prints. **[2]**

(d) Rewrite the loop so that it also counts the days with more than 20 mm of rain. **[3]**

Question 2

A school wants to store the results of a quiz in a two-dimensional array. Each row holds the name of a student followed by the scores of five rounds.

(a) Describe how a two-dimensional array is stored and accessed. **[3]**

(b) A student’s scores are 8, 6, 9, 7 and 10.

(i) Calculate the mean score. **[1]**

(ii) Write a Python function **mean(scores)** that returns the mean of a list of numbers. **[3]**

(c) Give one reason why a programmer might use a function rather than repeating code. **[2]**

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Section B

Question 3

The following algorithm is intended to search a sorted list for a target value using a binary search. It contains one logic error.

```
FUNCTION search(items, target)
    low <- 0
    high <- LEN(items) - 1
    WHILE low <= high
        mid <- (low + high) DIV 2
        IF items[mid] = target THEN
            RETURN mid
        ELSE IF items[mid] < target THEN
            high <- mid - 1
        ELSE
            low <- mid + 1
        ENDIF
    ENDWHILE
    RETURN -1
ENDFUNCTION
```

(a) Complete the trace table for the call search([3, 8, 11, 15, 21, 30], 21). **[4]**

low	high	mid	items[mid]
0	5		

(b) Identify the logic error and write the corrected lines. **[2]**

(c) State the maximum number of comparisons needed to find a value in a sorted list of 1,000 items. **[1]**

Question 4

A library lends laptops to students for a maximum of three days. Design an algorithm that reads the date a laptop was borrowed and the date it was returned, and outputs the fine due if it is late.

(a) The fine is 50 pence for each day late, up to a maximum of £5.00. Write your algorithm in pseudocode or Python. **[6]**

(b) Suggest two items of test data for your algorithm and give the expected result for each. **[4]**

END OF QUESTIONS

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